
A Step in the Right Direction: Combining Point Forecasts with Foundation-Model Signals

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Abstract

1 Point forecast accuracy typically deteriorates as the forecast horizon increases, but
2 considerably less is known about how directional accuracy varies across horizons.
3 In many operational and financial settings, the direction of change is itself decision
4 relevant. Recent time series foundation models offer new opportunities to
5 study directional prediction. We introduce a Directional Prediction Framework.
6 It reformulates forecasting datasets as directional prediction tasks. Using directional
7 accuracy as a common evaluation criterion, the framework enables direct
8 comparison between point forecasting methods and time series classifiers. On
9 the M4 Daily subset, Mantis exhibits higher directional accuracy at some longer
10 forecast horizons. We further present a post hoc proof of concept analysis in which
11 directional adjustment is associated with lower point forecast error.

12 1 Introduction

13 Point forecast accuracy typically deteriorates as the forecast horizon increases (Hyndman and Athana-
14 sopoulos, 2021). However, relatively little is known about how directional accuracy varies across
15 horizons. This gap raises a natural question: *does directional accuracy exhibit the same horizon*
16 *dependent behaviour as point forecast accuracy?*

17 In many applications, decisions depend less on the exact future value of a quantity than on whether it
18 will increase or not increase. Although directional accuracy has long been recognised as an important
19 dimension of forecast evaluation (Leitch and Tanner, 1991; Granger and Pesaran, 2000), direction
20 is typically evaluated as a by product of point forecasts rather than modelled as a prediction target.
21 Forecasting datasets are not structured for directional prediction, complicating comparisons between
22 point forecasts and classifier predictions. Forecast combinations can improve accuracy (Bates and
23 Granger, 1969), but whether signals from different model families can do the same remains unclear.

24 To investigate this possibility, we propose a Directional Prediction Framework that reformulates
25 forecasting datasets into time series classification tasks with directional labels. Using directional
26 accuracy as a common evaluation criterion, the framework enables direct comparison between point
27 forecasting methods and time series classifiers, including foundation model based approaches. With
28 the M4 dataset, we descriptively investigate how directional accuracy evolves across forecast horizons.
29 We then conduct a post hoc proof of concept analysis of whether learned directional signals can
30 improve point forecast accuracy through directional adjustment.

31 Our results suggest that directional accuracy does not necessarily follow the same horizon dependent
32 pattern as point forecast accuracy. On the M4 Daily subset, Mantis exhibits higher directional accuracy
33 at some longer forecast horizons. Among the evaluated frequencies, Daily exhibits the largest
34 observed directional adjustment improvement, although its magnitude depends on the forecasting
35 setting. Together, these findings suggest that directional prediction merits investigation as a task in its
36 own right.

Key contributions include: **(i)** We introduce a Directional Prediction Framework that reformulates forecasting datasets as time series classification tasks with directional labels. Directional accuracy provides a common criterion for directly comparing point forecasting methods and time series classifiers, including foundation model based approaches; **(ii)** We document descriptive horizon dependent differences in directional accuracy. On the M4 Daily subset, Mantis exhibits higher directional accuracy at some longer forecast horizons; and **(iii)** We provide post hoc proof of concept evidence that learned directional signals can complement point forecasts through directional adjustment.

2 Related Work

Directional accuracy measures whether a prediction correctly identifies the direction of change of a target variable. It has long been recognised as an important dimension of forecast evaluation because many decisions depend more on anticipating future movements than on minimising point forecast error (Leitch and Tanner, 1991; Granger and Pesaran, 2000). Despite its practical importance, directional prediction has traditionally been treated as a secondary outcome of point forecasting rather than as a task in its own right (De Gooijer and Hyndman, 2006; Hyndman and Koehler, 2006). Recent studies underscore the need for precise directional evaluation protocols, as ambiguous formulations can distort directional accuracy estimates (McCarthy and Snudden, 2025).

Recent advances in **time series classification** (TSC) have produced effective methods for extracting or learning discriminative temporal representations, including ROCKET and InceptionTime, together with standardised evaluation frameworks based on benchmark repositories such as the UCR Archive (Bagnall et al., 2017; Dau et al., 2019; Dempster et al., 2020; Ismail Fawaz et al., 2020). At the same time, **time series foundation models** have created new opportunities for forecasting and classification. Chronos-2 uses group attention for zero-shot univariate, multivariate, and covariate informed forecasting (Ansari et al., 2025). Mantis is a lightweight transformer based foundation model for time series classification trained with self supervised contrastive learning (Feofanov et al., 2025).

These developments motivate a unified view of directional prediction. Although point forecasting methods and time series classifiers, including foundation model based approaches, all capture temporal information, they are typically evaluated using different criteria and therefore remain difficult to compare directly. Our framework enables their comparison using directional accuracy as a common evaluation criterion across forecast horizons.

3 Method

We introduce a Directional Prediction Framework that reformulates each forecasting task as a binary classification task. Whether a future observation increases relative to the forecast origin. The framework evaluates point forecasting methods and time series classifiers using directional accuracy. When classifier predicted and forecast implied labels disagree, the classifier predicted label determines which point forecast adjustment factor is selected.

Forecasting notation. Let $\mathbf{y} = (y_1, y_2, \dots, y_T)$ be the observed time series, with forecast origin T and final observed value y_T . At horizon h , y_{T+h} is the future observation and $\hat{y}_{T+h|T}$ is its point forecast. Let H denote the maximum forecast horizon, with $h \in \{1, \dots, H\}$.

Binary sign convention. Following McCarthy and Snudden (2025), define $\text{sign}(y) = 1$ if $y > 0$, and $\text{sign}(y) = -1$ if $y \leq 0$.

Directional accuracy. At forecast origin T and horizon h , directional accuracy is defined as

$$d_{T,h} = \mathbb{I}[\text{sign}(y_{T+h} - y_T) = \text{sign}(\hat{y}_{T+h|T} - y_T)]. \quad (1)$$

Both changes are evaluated relative to the forecast origin y_T . For a collection of N time series, let $i \in \{1, \dots, N\}$ index the individual series. Across series directional accuracy is

$$\text{DA}_{T,h} = \frac{1}{N} \sum_{i=1}^N d_{i,T,h}, \text{ the mean of the series level indicators (Appendix C).}$$

Directional class labels. At forecast origin T and horizon h , define

$$z_{T,h} = \mathbb{I}[\text{sign}(y_{T+h} - y_T) = 1], \quad (2)$$

where $z_{T,h} = 1$ denotes an increase from y_T to y_{T+h} , and $z_{T,h} = 0$ denotes a non increase.

85 **Directional Prediction Framework.** The framework proceeds in four steps (Figure 1).

86 **1. Classifier predicted directional class label.** At forecast origin T and horizon h , a time series

87 classifier produces $\hat{z}_{T,h} \in \{0, 1\}$, where $\hat{z}_{T,h} = 1$ denotes a predicted increase relative to y_T , and

88 $\hat{z}_{T,h} = 0$ denotes a predicted non increase.

89 **2. Forecast implied directional class label.** The point forecast implies

$$\hat{z}_{T,h}^F = \mathbb{I} [\text{sign}(\hat{y}_{T+h|T} - y_T) = 1]. \quad (3)$$

90 **3. Directional adjustment.** The adjustment factor is

$$\gamma_{T,h} = \begin{cases} 1, & \hat{z}_{T,h} = \hat{z}_{T,h}^F, \\ \lambda_{\text{Up}}, & \hat{z}_{T,h} \neq \hat{z}_{T,h}^F \wedge \hat{z}_{T,h} = 1, \\ \lambda_{\text{Down}}, & \hat{z}_{T,h} \neq \hat{z}_{T,h}^F \wedge \hat{z}_{T,h} = 0. \end{cases} \quad (4)$$

91 where $\lambda_{\text{Up}} \geq 1$ and $0 < \lambda_{\text{Down}} \leq 1$ are multiplicative adjustment parameters.

92 **4. Adjusted forecast.** The adjusted point forecast is

$$\tilde{y}_{T+h|T} = \gamma_{T,h} \hat{y}_{T+h|T}. \quad (5)$$

93 The adjustment in equation (5) is used as a post hoc exploratory proof of concept to assess whether

94 learned directional information can improve point forecast accuracy, rather than as an optimal

95 adjustment rule. We use a multiplicative adjustment because M4 target values are positive.

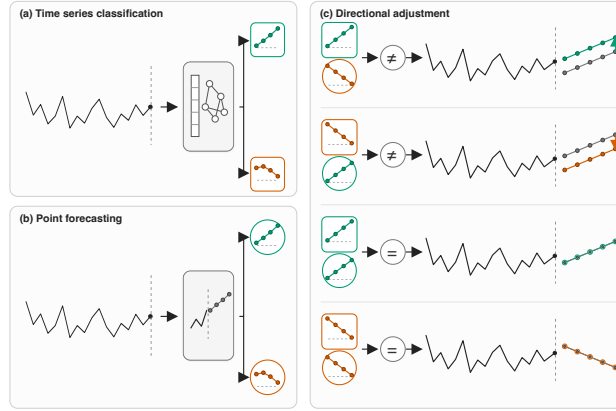


Figure 1: The Directional Prediction Framework. Squares and circles denote classifier predicted and forecast implied labels; disagreement determines which adjustment factor is selected.

96 4 Experiments

97 We evaluate the Directional Prediction Framework on the M4 dataset (Makridakis et al., 2018),

98 comprising 100,000 time series across six frequencies and multiple forecast horizons. Following

99 Section 3, we reformulate the forecasting tasks as directional prediction tasks. We evaluate point

100 forecasting methods and time series classifiers, including foundation model based approaches, using

101 directional accuracy. We compare the M4 winning SMYL (Smyl, 2020) and second ranked FFORMA

102 (Montero-Manso et al., 2020) point forecasts with Chronos-2 (Ansari et al., 2025). The classifier

103 comparison includes 1-NN DTW, 1-NN ED, Rotation Forest, XGBoost, ROCKET (Dempster et al.,

104 2020), InceptionTime (Ismail Fawaz et al., 2020), and Mantis (Feofanov et al., 2025). The con-

105 ventional classifiers span distance based, feature based, ensemble, kernel based, and deep learning

106 approaches. Chronos-2 uses released weights without fine tuning but has documented M4 pretraining

107 overlap. Mantis uses a frozen encoder with a 200 tree Random Forest. Implementation and dataset

108 construction details are provided in Appendices C–D. Directional prediction is evaluated indepen-

109 dently at each horizon. Tables 1 and 2 report directional accuracy at the terminal official horizon

110 $h = H$, whereas sMAPE, MASE, and OWA are computed over the complete official forecast horizon.

111 Forecast implied directional labels are calculated using equation (3). We assess whether learned

112 directional signals improve point forecast accuracy through a post hoc parameter grid analysis on the

113 official M4 evaluation set (Appendix B). Accordingly, adjusted results are interpreted as exploratory

114 proof of concept estimates rather than held out performance estimates.

5 Results

Figure 2 descriptively shows how directional accuracy varies across horizons on the M4 Daily subset, with the highest observed values occurring at some longer horizons.

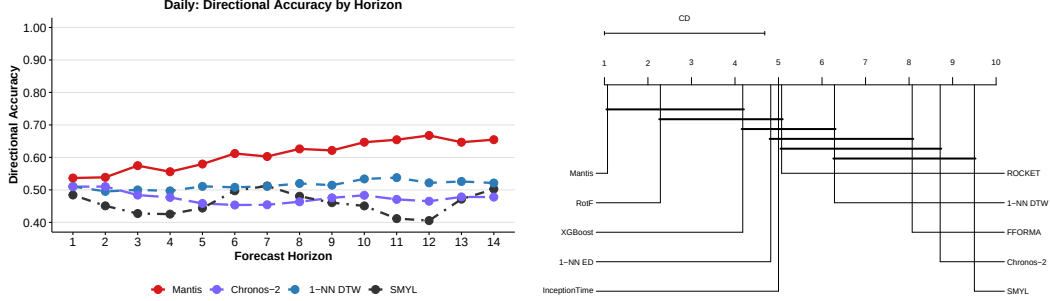


Figure 2: Directional prediction results for the M4 Daily subset. **Left:** Directional accuracy by forecast horizon. **Right:** Descriptive average ranks across Daily forecast horizons.

Mantis achieves the best average rank across the Daily forecast horizons. This result is not driven solely by an individual horizon. Across the remaining frequencies, directional profiles are heterogeneous, and no single method dominates (Appendix A). Reproducibility details can be read in Appendix D.

Table 1 reports the forecasting value of directional adjustment on the Daily subset. Relative to their corresponding unadjusted forecasts, the Daily OWA reductions are 4.01% for SMYL–Mantis and 5.52% for Chronos-2–Mantis. The 7.64% value in the table instead compares Chronos-2–Mantis with SMYL. The infeasible SMYL–Oracle achieves a substantially larger 47.34% OWA improvement relative to SMYL, indicating considerable ex post headroom within the scaling construction rather than the performance of a feasible forecasting method.

Table 1: Daily frequency results. DA is evaluated at the terminal horizon $h = 14$; sMAPE, MASE, and OWA are computed over all 14 official forecast horizons. The best non oracle value in each column is shown in bold.

Method	sMAPE↓	MASE↓	DA↑	OWA↓	Improvement over SMYL (%)		
					sMAPE↑	MASE↑	OWA↑
Chronos-2	3.09	3.38	0.527	1.023	2.61	1.88	2.24
FFORMA	3.10	3.34	0.510	1.019	2.29	2.95	2.62
SMYL	3.17	3.45	0.502	1.046	0.00	0.00	0.00
SMYL–Mantis	3.05	3.30	0.670	1.004	3.69	4.33	4.01
Chronos-2–Mantis	2.95	3.16	0.665	0.966	6.86	8.41	7.64
SMYL–Oracle	1.72	1.76	0.859	0.551	45.64	49.02	47.34

6 Conclusion

We introduced a Directional Prediction Framework that enables point forecasting methods and time series classifiers to be evaluated using directional accuracy and uses learned directional signals to adjust point forecasts. On the M4 Daily subset, Mantis exhibits higher directional accuracy at some longer horizons, while directional accuracy profiles vary across frequencies. Mantis based directional adjustment also lowers the OWA of SMYL and Chronos-2 on the Daily subset. Gains vary across frequencies and forecasting settings. These findings provide exploratory proof of concept evidence that learned directional representations can complement point forecasts. Time series foundation models therefore offer opportunities to study directional prediction as a complementary forecasting task. The evidence is limited to the M4 dataset and the methods considered. Future work should evaluate additional datasets and representations, investigate mechanisms underlying directional predictability, and develop more adaptive adjustment strategies. All comparisons are descriptive and conditional on the reported experimental design (Appendix D).

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191 A Forecasting Value of Directional Information

192 This appendix examines whether learned directional signals provide additional value for point
 193 forecasting across the six M4 frequencies. Figure 3 shows two examples of directional adjustment
 194 when classifier predicted and forecast implied directions disagree.

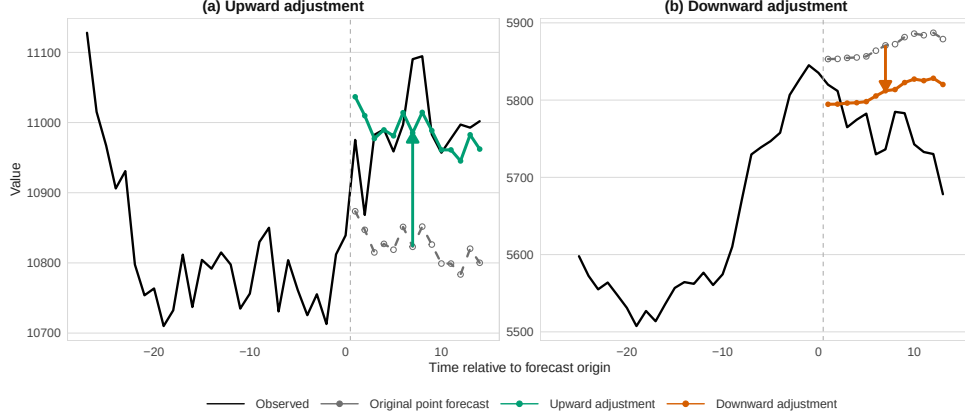


Figure 3: Examples of directional adjustment. Mantis adjusts the SMYL forecast upward for M4 Daily series D1602 (left) and the Chronos-2 forecast downward for M4 Weekly series W39 (right).

Table 2: Directional accuracy at the terminal official horizon and point-forecast accuracy over the complete official forecast horizon, by M4 frequency.

Panel A. Directional accuracy at $h = H$ (DA) $\uparrow$

Method	Hourly	Daily	Weekly	Monthly	Quarterly	Yearly	All
Chronos-2	0.802	0.527	0.705	0.664	0.657	0.676	0.660
FFORMA	0.688	0.510	0.649	0.694	0.706	0.765	0.705
SMYL	0.761	0.502	0.635	0.722	0.717	0.784	0.726
SMYL–Mantis	0.766	0.670	0.652	0.722	0.721	0.786	0.734
Chronos-2–Mantis	0.809	0.665	0.708	0.665	0.693	0.725	0.686
Mantis	0.652	0.658	0.593	0.521	0.699	0.778	0.629
SMYL–Oracle	0.829	0.859	0.841	0.859	0.861	0.909	0.871

Panel B. Overall Weighted Average (OWA) $\downarrow$

Method	Hourly	Daily	Weekly	Monthly	Quarterly	Yearly	All
Chronos-2	0.393	1.023	0.712	0.888	0.911	0.891	0.895
FFORMA	0.484	1.019	0.796	0.858	0.847	0.799	0.838
SMYL	0.440	1.046	0.851	0.836	0.847	0.778	0.821
SMYL–Mantis	0.438	1.004	0.839	0.836	0.847	0.778	0.819
Chronos-2–Mantis	0.390	0.966	0.695	0.888	0.898	0.864	0.880
SMYL–Oracle	0.356	0.551	0.540	0.557	0.502	0.383	0.479

Panel C. OWA improvement relative to SMYL (%) $\uparrow$

Method	Hourly	Daily	Weekly	Monthly	Quarterly	Yearly	All
Chronos-2	10.78	2.24	16.29	-6.29	-7.57	-14.51	-9.01
FFORMA	-9.92	2.62	6.49	-2.65	0.03	-2.69	-2.05
SMYL	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SMYL–Mantis	0.57	4.01	1.42	0.00	0.07	0.02	0.25
Chronos-2–Mantis	11.31	7.64	18.27	-6.23	-6.00	-11.02	-7.19
SMYL–Oracle	19.03	47.34	36.52	33.40	40.69	50.76	41.61

Notes.

- (i) Mantis appears only in Panel A; Panels B–C report point-forecasting methods.
- (ii) Bold marks the best non-oracle value in each column.
- (iii) Panel C reports $100(1 - \text{OWA}_m / \text{OWA}_{\text{SMYL}})$ from unrounded values; Panel B displays three decimals.
- (iv) Panel A uses terminal horizons ($H = 48, 14, 13, 18, 8, 6$) for Hourly–Yearly, respectively; Panels B–C use the complete official horizons. For All, sMAPE, MASE, and DA are weighted by frequency series counts, and OWA is recomputed from the aggregated sMAPE and MASE values relative to Naive2.
- (v) SMYL–Oracle is an infeasible ex post diagnostic.

Table 2 reports terminal horizon directional accuracy and full horizon point forecast accuracy. The Daily subset offers the clearest demonstration. Mantis based adjustment increases Chronos-2 terminal horizon directional accuracy from 0.527 to 0.665 and reduces OWA from 1.023 to 0.966. The same adjustment increases SMYL terminal horizon directional accuracy from 0.502 to 0.670 and reduces OWA from 1.046 to 1.004. These adjusted OWA results use the post hoc parameter grid analysis described in Appendix B and are therefore exploratory proof of concept estimates rather than held out performance estimates.

The Hourly and Weekly results extend this pattern beyond Daily data: Chronos-2–Mantis achieves the lowest non oracle OWA at all three frequencies. Comparable gains do not occur uniformly for the lower frequency subsets or in the aggregate results, indicating that the value of learned directional information is setting dependent.

Figures 2, 4, and 5 show the corresponding horizon structure. On the Daily subset, Mantis directional accuracy increases from 0.537 at the first horizon to 0.668 at horizon 12; the remaining frequencies exhibit more setting specific profiles. These descriptive horizon profiles vary across frequencies and need not follow the corresponding patterns of point forecast accuracy.

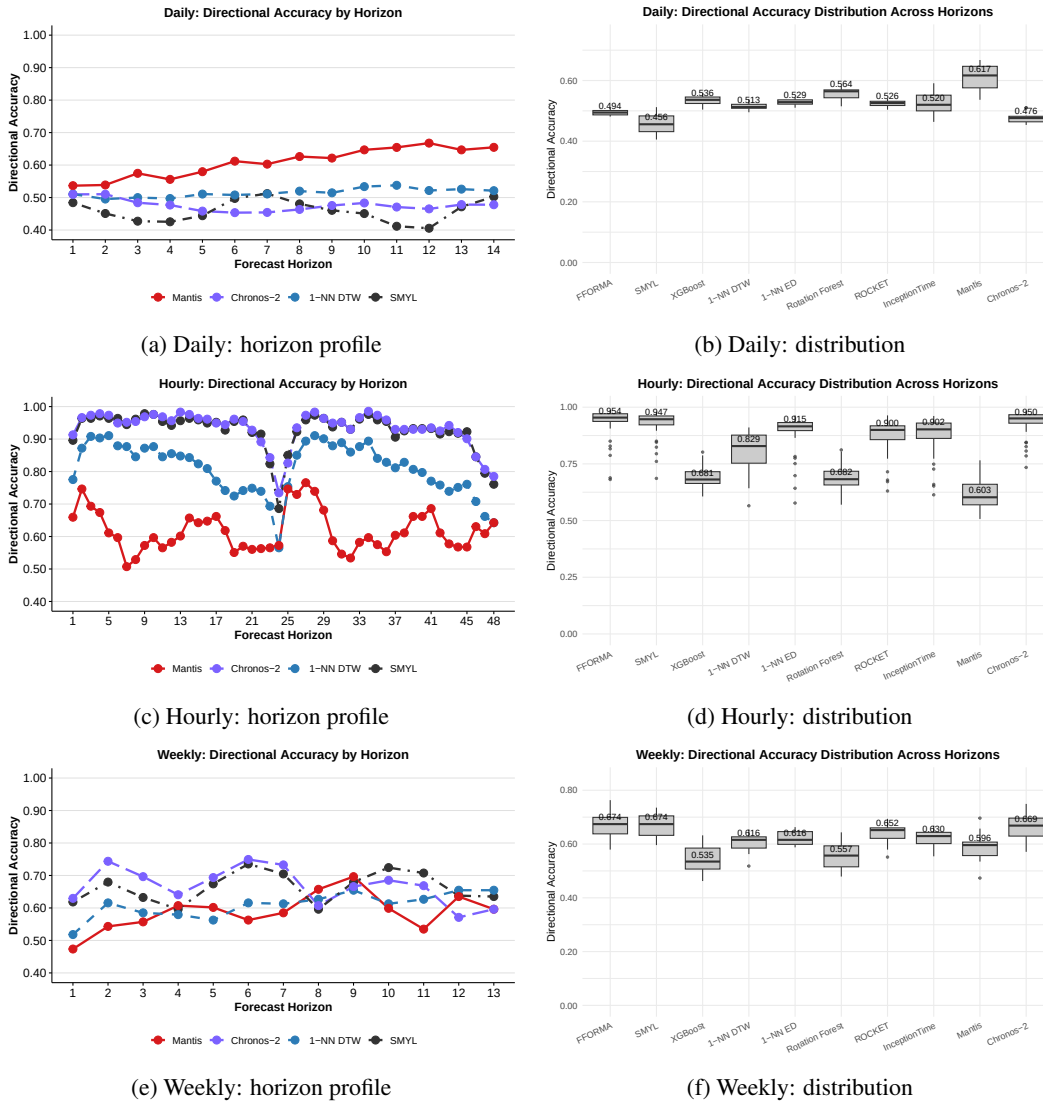


Figure 4: Directional accuracy across forecast horizons for the Daily, Hourly, and Weekly subsets of M4. **Left:** directional accuracy by forecast horizon for selected methods. **Right:** distribution of directional accuracy across horizons for all evaluated methods.

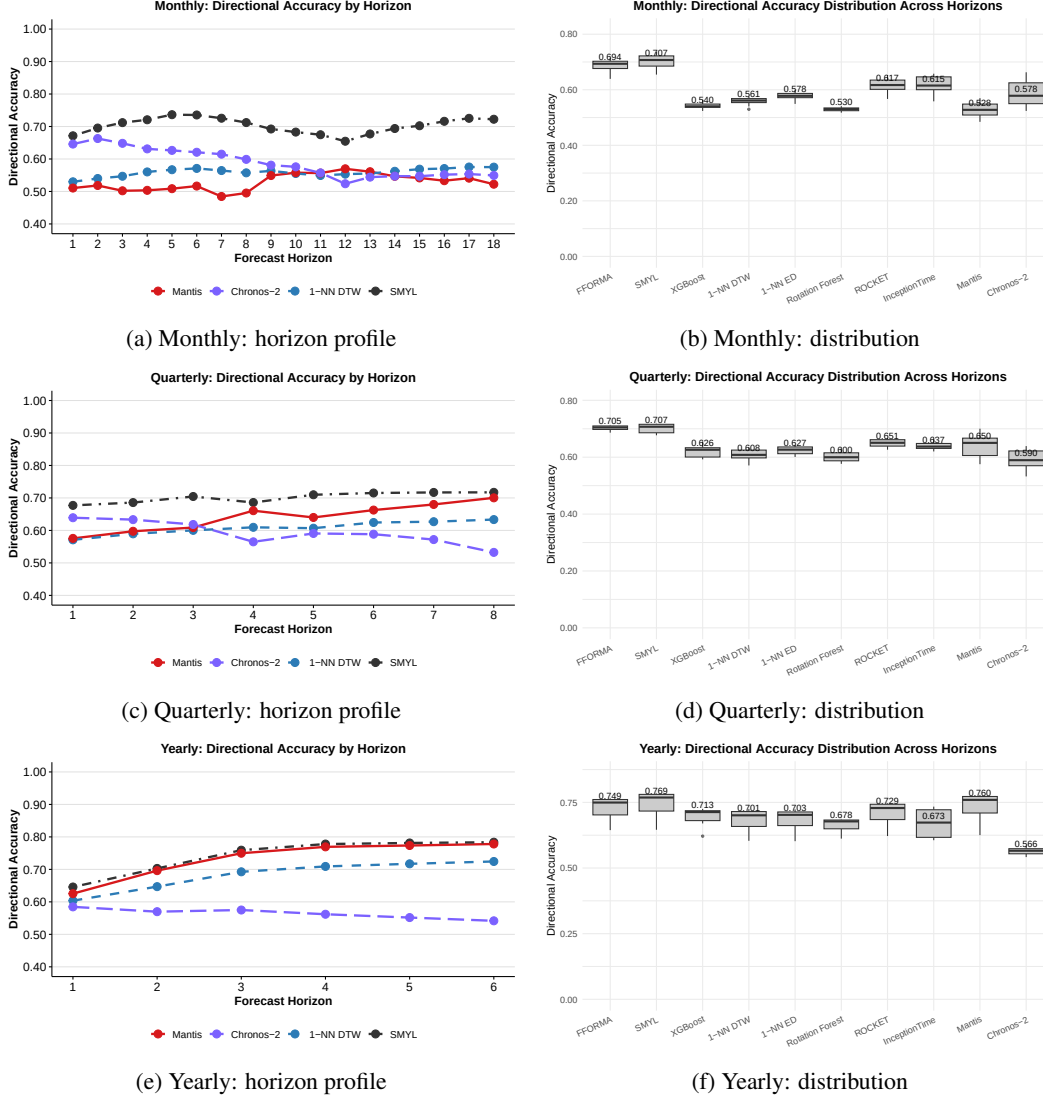


Figure 5: Directional accuracy across forecast horizons for the Monthly, Quarterly, and Yearly subsets of M4. **Left:** directional accuracy by forecast horizon for selected methods. **Right:** distribution of directional accuracy across horizons for all evaluated methods.

210 SMYL–Oracle is the infeasible ex post scaling diagnostic. For each series, it selects the multiplier
 211 $m \in \{0.500, 0.501, \dots, 1.500\}$ that minimises SMYL sMAPE over that series realised official
 212 forecast horizon. On the Daily subset, it improves OWA by 47.34% relative to SMYL and by 4.01%
 213 relative to SMYL–Mantis. This difference indicates headroom within the scaling construction when
 214 realised outcomes are available, but it should not be attributed solely to unexploited directional
 215 information.

216 Together, the results provide exploratory evidence that learned directional representations can com-
 217 plement point forecasts in suitable settings. The Directional Prediction Framework makes this
 218 contribution measurable across point forecasting methods, time series classifiers, and future founda-
 219 tion model representations.

220 B Sensitivity Analysis of Directional Adjustment

221 This appendix assesses whether directional adjustment gains are robust to variation in the adjustment
 222 parameters. For each model frequency combination, we evaluated 525 global parameter pairs. With
 223 $\lambda_{Up} \in [1.00, 1.12]$ and $\lambda_{Down} \in [0.90, 1.00]$, in increments of 0.005. We evaluated each pair using
 224 OWA on the official M4 evaluation set and report the pair that minimised OWA. Consequently, these
 225 adjusted results are exploratory proof of concept estimates rather than held out performance estimates.

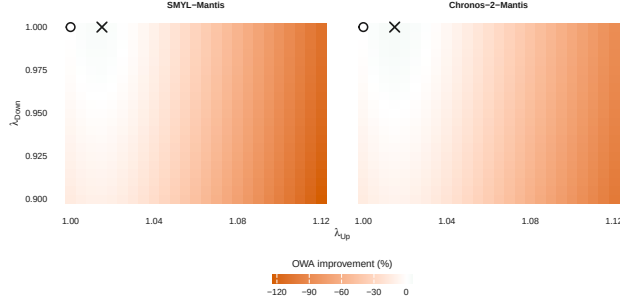


Figure 6: Directional adjustment sensitivity for the Daily frequency. Circles denote the no adjustment setting $(\lambda_{Up}, \lambda_{Down}) = (1, 1)$; crosses denote the lowest OWA parameter pair in each panel.

226 Figure 6 illustrates that the daily response surfaces are smooth in the vicinity of the best-performing
 227 settings. Performance varies more strongly with λ_{Up} than with λ_{Down} , but improvements occur over
 228 local regions rather than at isolated parameter pairs.

Table 3: Summary of directional-adjustment sensitivity across M4 frequencies.

Panel A. Best-performing parameter pair						
Frequency	Adjusted method	λ_{Up}^*	λ_{Down}^*	Base OWA↓	Best OWA↓	Best improvement (%)↑
Hourly	Chronos-2-Mantis	1.005	1.000	0.393	0.390	0.60
Hourly	SMYL-Mantis	1.005	1.000	0.440	0.438	0.57
Daily	Chronos-2-Mantis	1.015	1.000	1.023	0.966	5.52
Daily	SMYL-Mantis	1.015	1.000	1.046	1.004	4.01
Weekly	Chronos-2-Mantis	1.015	0.990	0.712	0.695	2.38
Weekly	SMYL-Mantis	1.015	0.995	0.851	0.839	1.42
Monthly	Chronos-2-Mantis	1.005	1.000	0.888	0.888	0.06
Monthly	SMYL-Mantis	1.000	1.000	0.836	0.836	0.00
Quarterly	Chronos-2-Mantis	1.025	1.000	0.911	0.898	1.45
Quarterly	SMYL-Mantis	1.005	1.000	0.847	0.847	0.07
Yearly	Chronos-2-Mantis	1.065	0.995	0.891	0.864	3.05
Yearly	SMYL-Mantis	1.010	1.000	0.778	0.778	0.02

Panel B. Robustness across the parameter grid				
Frequency	Adjusted method	λ_{Up} range	λ_{Down} range	Improving surface (%)↑
Hourly	Chronos-2-Mantis	1.000–1.005	1.000–1.000	0.19
Hourly	SMYL-Mantis	1.000–1.010	0.995–1.000	0.38
Daily	Chronos-2-Mantis	1.010–1.020	0.985–1.000	9.33
Daily	SMYL-Mantis	1.010–1.015	0.985–1.000	6.67
Weekly	Chronos-2-Mantis	1.005–1.025	0.980–1.000	7.81
Weekly	SMYL-Mantis	1.005–1.035	0.985–1.000	6.48
Monthly	Chronos-2-Mantis	1.000–1.015	0.975–1.000	0.38
Monthly	SMYL-Mantis	1.000–1.010	0.980–1.000	0.00
Quarterly	Chronos-2-Mantis	1.005–1.045	0.900–1.000	35.43
Quarterly	SMYL-Mantis	1.000–1.030	0.910–1.000	0.95
Yearly	Chronos-2-Mantis	1.030–1.110	0.905–1.000	94.48
Yearly	SMYL-Mantis	1.000–1.085	0.915–1.000	1.14

Notes.

- (i) Panel A reports the lowest OWA observed over the evaluated parameter grid and its corresponding parameter pair.
- (ii) Improving surface is the percentage of evaluated parameter pairs that produce lower OWA than the corresponding unadjusted forecast.
- (iii) OWA values are displayed to three decimals; percentage improvements are calculated from unrounded values.
- (iv) The parameter ranges report the marginal minimum and maximum values of each parameter among evaluated pairs producing OWA within 1% of the best value; they do not imply that every combination within the resulting rectangle satisfies this criterion.

Table 3 reports OWA improvement relative to each method’s corresponding unadjusted base forecast. For Daily, the reductions are 4.01% for SMYL–Mantis relative to SMYL and 5.52% for Chronos-2–Mantis relative to Chronos-2. Daily shows the largest within model improvements, whereas Monthly adjustments provide negligible gains. Quarterly and Yearly Chronos-2–Mantis adjustments have the broadest improving regions but more modest best improvements. Thus, attainable gain and the extent of the parameter region over which improvement occurs are distinct properties.

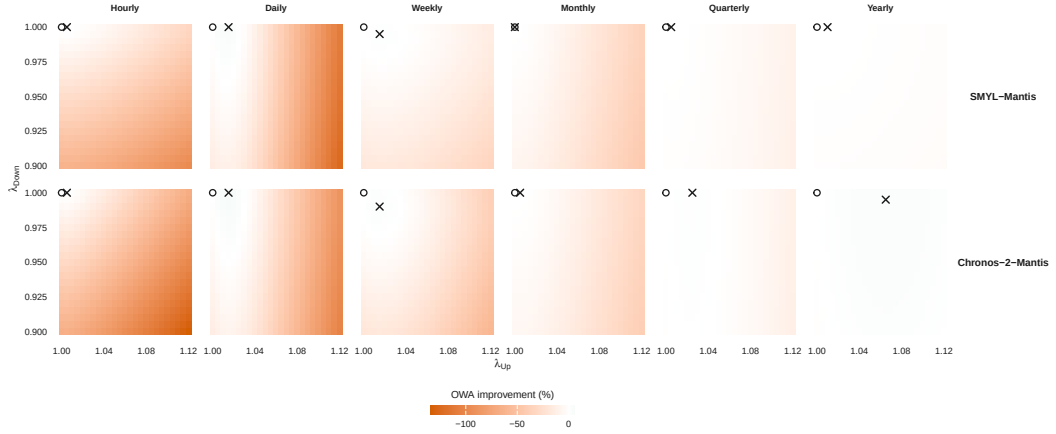


Figure 7: Directional adjustment sensitivity across M4 frequencies and base forecasting methods. Circles denote the no adjustment setting; crosses denote the lowest OWA parameter pair in each panel.

Figure 7 illustrates that the shape and extent of the improving regions differ across frequencies and base forecasts. Surfaces remain smooth throughout the evaluated grid. At the selected Daily setting $(\lambda_{Up}, \lambda_{Down}) = (1.015, 1.000)$. Mantis disagrees with SMYL and Chronos-2 on 46.94% and 38.06% of the series, respectively. Among these disagreements, the adjusted terminal forecast adopts the Mantis label in 86.90% and 76.88% of cases, respectively, while OWA decreases by 8.48% and 13.31% within the corresponding disagreement subsets. Because $\lambda_{Down} = 1$, forecasts associated with downward disagreements remain unchanged.

C Directional Prediction Dataset Construction

Section 3 defines the directional labels and evaluation criterion. Forecasting datasets such as M4 are not naturally structured as directional prediction tasks. This appendix operationalises the framework by converting unequal length M4 series into fixed length, labelled instances for horizon specific binary classification. The resulting datasets follow the labelled series conventions used in the UCR Archive and TSC Bake Off studies (Dau et al., 2019; Bagnall et al., 2017). Point forecasting methods retain their native forecast generation procedures.

Directional accuracy across series. For a method applied to a dataset containing N time series, let $i \in \{1, \dots, N\}$ denote the index of an individual series. Directional accuracy at forecast origin T and horizon h is defined as

$$DA_{T,h} = \frac{1}{N} \sum_{i=1}^N d_{i,T,h}, \quad (6)$$

where $d_{i,T,h}$ denotes the directional accuracy indicator in equation (1) for time series i .

Construction of directional prediction tasks. Consider one rolling window of length w , ending at forecast origin T :

$$\mathbf{y}_{T,w} = (y_{T-w+1}, \dots, y_T).$$

The procedure is repeated across rolling windows and time series.

1. **Construct rolling windows.** Following the rolling origin principle described by Hyndman and Athanasopoulos (2021), fixed length windows are generated using window length w , maximum forecast horizon H , and step size $w + H$. For the reported results, $w = (256, 64, 64, 64, 32, 16)$ for Hourly, Daily, Weekly, Monthly, Quarterly, and Yearly series, respectively.
2. **Standardise the rolling windows.** Let μ and σ be the mean and standard deviation of $\mathbf{y}_{T,w}$. If $\sigma > 0$, each observation is standardised as

$$y_t^{\text{std}} = \frac{y_t - \mu}{\sigma}, \quad t = T - w + 1, \dots, T.$$

Applying this transformation elementwise gives the standardised window $\mathbf{y}_{T,w}^{\text{std}}$, which is the classifier input. If $\sigma = 0$, $\mathbf{y}_{T,w}^{\text{std}}$ is set to the zero vector.

3. **Assign directional class labels.** For each forecast horizon $h = 1, \dots, H$, the window is assigned the directional label $z_{T,h}$ defined in equation (2). The future observation y_{T+h} is used only to assign the label and is not included in the classifier input.
4. **Fit the classifiers.** Series yielding at least one complete length w input window followed by H outcomes were randomly assigned to training and internal hold out partitions using an 80:20 series level split. Ineligible series were omitted from classifier training without padding or imputation, and all windows from a series remained in the same partition. A separate classifier was fitted for each horizon using only windows from the training series.

This construction converts unequal length forecasting series into fixed length, scale standardised, horizon specific directional prediction datasets while preserving the information available at each forecast origin. It thereby enables time series classifiers and point forecasting methods to be evaluated on a common directional task. Figure 8 summarises the rolling window construction and resulting time series classification tasks.

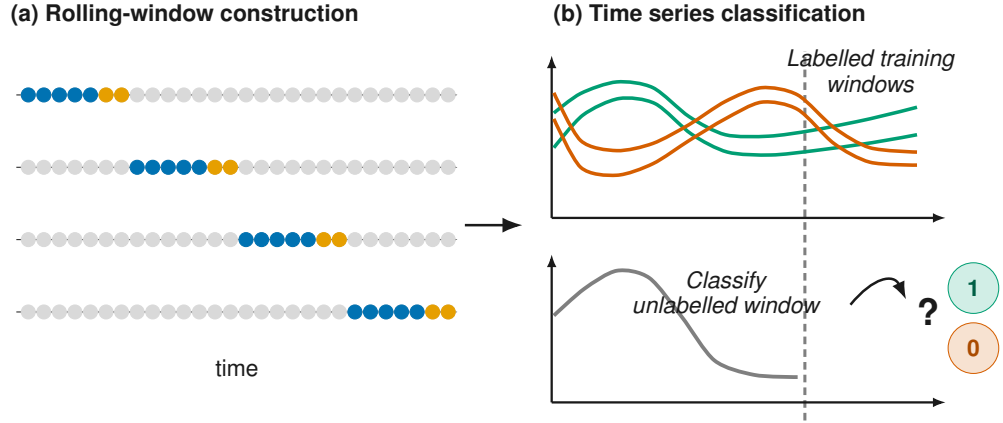


Figure 8: Construction of horizon-specific directional prediction datasets. **(a)** Blue points denote fixed-length classifier input windows, orange points denote subsequent observations used to assign directional labels, and grey points lie outside the corresponding instance. **(b)** For a given horizon, labelled windows define a binary time-series classification task in which a new window is assigned class 0 or 1.

277 D Experimental Design and Reproducibility

278 **Dataset characteristics.** Experiments use the M4 dataset (Makridakis et al., 2018). Table 4 reports
 279 the number of series by frequency and domain. Figure 9 reports directional class composition at each
 280 frequency’s terminal official forecast horizon $h = H$, using the labels defined in equation (2).

Table 4: Number of M4 series by frequency and domain.

Frequency	Micro	Industry	Macro	Finance	Demographic	Other	Total
Yearly	6,538	3,716	3,903	6,519	1,088	1,236	23,000
Quarterly	6,020	4,637	5,315	5,305	1,858	865	24,000
Monthly	10,975	10,017	10,016	10,987	5,728	277	48,000
Weekly	112	6	41	164	24	12	359
Daily	1,476	422	127	1,559	10	633	4,227
Hourly	0	0	0	0	0	414	414
Total	25,121	18,798	19,402	24,534	8,708	3,437	100,000

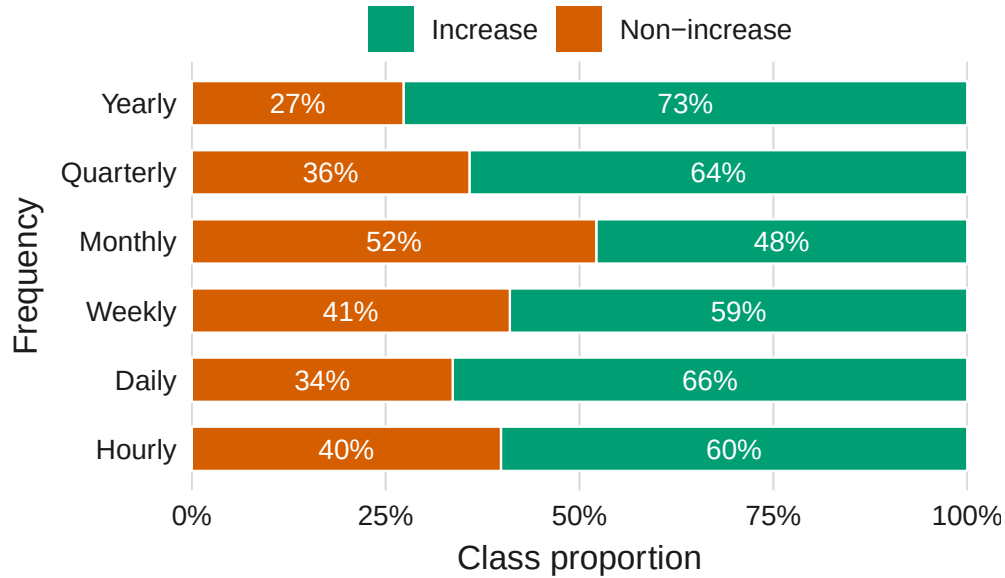


Figure 9: Directional class proportions at each M4 frequency’s terminal official forecast horizon $h = H$, calculated across all series within that frequency. Increase denotes $y_{T+h} > y_T$; non increase denotes $y_{T+h} \leq y_T$, where y_T is the final observed value of each series.

281 **Training and evaluation protocol.** Dataset construction, including the series level split, training
 282 window eligibility, and classifier fitting, is described in Appendix C. At each horizon, every M4
 283 series contributed one directional prediction at its official final forecast origin; these all series results
 284 include both training partition and held out series, although the corresponding future observations
 285 were not used for classifier training. Directional accuracy was computed using equations (1) and (6).

286 **Model configuration.** SMYL and FFORMA forecasts were obtained from M4comp2018 (Montero-
 287 Manso et al., 2018). Chronos-2 used the publicly available amazon/chronos-2 checkpoint without
 288 task specific fine tuning. Its pretraining data include the M4 Daily, Hourly, Monthly, and Weekly
 289 subsets (Ansari et al., 2025); results for these frequencies therefore involve pretraining overlap.
 290 Mantis followed the authors’s frozen backbone feature extraction protocol (Feofanov et al., 2025):
 291 representations were extracted without fine tuning and classified using a 200 tree Random Forest
 292 with unrestricted maximum depth. The pretraining corpus reported for Mantis does not include M4;
 293 therefore, unlike Chronos-2, Mantis has no documented direct M4 pretraining overlap. ROCKET,

294 InceptionTime, 1-NN DTW, 1-NN ED, and Rotation Forest used their standard `sktime` implemen-
295 tations (Löning et al., 2019). For each frequency, window setting, and horizon, XGBoost used a
296 binary logistic objective; its eight hyperparameters were selected by five fold log loss Bayesian
297 optimisation with 8 initial and 20 sequential evaluations, at most 1,500 boosting rounds, and 15 round
298 early stopping, following the FFORMA tuning approach (Montero-Manso et al., 2020). Directional
299 adjustment follows equations (4) and (5), using the post hoc parameter values reported in Appendix B.

300 **Model comparison.** The Daily critical difference diagram compares methods with complete horizon
301 coverage and was generated using `scmamp:plotCD` with $\alpha = 0.05$ (Calvo and Santafé, 2016). The
302 right panel of Figure 2 reports descriptive average ranks across the Daily forecast horizons.

303 **Compute resources.** The experiments were executed primarily on a local workstation running
304 Ubuntu 24.04 LTS, equipped with an AMD Ryzen 9 3950X CPU (16 cores and 32 threads), 128 GB
305 of RAM, and one NVIDIA GeForce RTX 5090 GPU with 32 GB of VRAM. Exact wall-clock times
306 and the additional compute consumed by preliminary experiments were not systematically recorded;
307 runtimes may also vary with software configuration and available parallelism.

308 **Reproducibility.** An anonymized repository accompanying the submission contains the complete
309 R–Python pipeline for dataset construction, fixed series-level splits, model fitting and evaluation,
310 Chronos-2 inference, sensitivity analysis, and regeneration of the reported tables and figures. The
311 scripts record the seeds, window settings, model hyperparameters, checkpoint identifiers, and adjust-
312 ment grid.

313 The README documents installation and the exact execution order. Versioned requirements, Conda
314 specifications, and Docker definitions specify the software environments, while generated artifacts
315 use fixed project paths.

316 **Asset licences.** The accompanying code developed for this study is released under the MIT License.
317 The M4 data are accessed through M4comp2018 version 0.2.0 (GPL-3); the `amazon/chronos-2`
318 and `paris-noah/Mantis-8M` checkpoints are released under Apache-2.0; `sktime` is released under
319 BSD-3-Clause; XGBoost is released under Apache-2.0; and `scmamp` is released under GPL-2. Exact
320 software versions and checkpoint identifiers are recorded in the accompanying repository. These
321 assets are credited to their original authors, used under their respective terms, and downloaded from
322 their official sources rather than redistributed.

323 **Code availability.** The anonymized repository is available to reviewers, and a cleaned public version
324 will be released after the review period.

325 D.1 Broader Impacts

326 Directional forecasting may improve decision support in domains where anticipating the direction
327 of change is more important than predicting its precise magnitude. However, incorrect directional
328 predictions or deployment beyond the evaluated M4 setting could lead to harmful financial or
329 operational decisions. Domain-specific validation, uncertainty assessment, monitoring, and human
330 oversight are therefore necessary, particularly in high-stakes applications.

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Answer: [No]

Justification: Point-forecast accuracy follows the official M4 evaluation protocol, with sMAPE, MASE, and OWA computed over the complete official forecast horizon, as described in Section 4 and Appendix A. These fixed-evaluation-set results are treated as descriptive; no error bars, confidence intervals, or fully specified inferential tests are reported for the main claims.

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